De Jong Yeong (DJ)

Software Engineer| Full-Stack Development | Applied Research

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LinkedIn: linkedin.com/in/de-jong-yeong | **GitHub:** github.com/dejongyeong | **Portfolio:** dejongyeong.com
Work Authorization: Stamp 4 Holder - Full right to work in Ireland. Open to relocation as required.

Professional Summary

Full Stack Engineer and AI Practitioner with experience building scalable web applications for connected machinery, real-time video streaming pipelines, and digital health platforms. Proficient across Next.js, TypeScript, Python, and SQL, with hands-on experience integrating computer vision workflows and a track record of resolving performance bottlenecks to reduce latency by 32%. Combines a pragmatic, resourceful mindset with the strong collaborative skills needed to engage with startup founders and key stakeholders, successfully aligning engineering outputs with business goals.

Technical Skills

Programming: TypeScript, Python, SQL, MongoDB, PHP; *Exposure to*: Java, C++, C#
Frameworks & Libraries: Next.js, REST APIs, React, PHP Laravel 8.0, Leaflet.js, Tailwind CSS, GStreamer
Data and Tooling: Git/GitHub, Vercel, Drizzle ORM, MongoDB, Supabase, GitHub Actions, Docker; *Exposure to:* GCP and AWS
Research Areas: Autonomous Systems, Multi-Sensor Fusion, Explainable AI (XAI)

Work Experience

Researcher, Full Stack Engineer

IMaR Technology Gateway | *2022 – present*

- Develop an industrial video management system (VMS) MVP for Liebherr Container Cranes by integrating RTSP/NDI streams into low-latency, picture-in-picture composite views utilising Python and GStreamer.
- Optimize multi-view video performance by designing custom Python-GStreamer pipelines that resolve synchronization and composition bottlenecks, reducing glass-to-glass latency by over 32% (340 ms to 230 ms) across 3-camera configurations.
- Lead bi-weekly software demonstrations to senior stakeholders, gathering iterative feedback to align cross-functional engineering and research outputs with client requirements.
- Developed a near-real-time agricultural telematics platform for Abbey Machinery, managing time-series data workflows and interactive data visualisations utilising Next.js, InfluxDB, Supabase, Python, Leaflet.js, and Ant Design.
- Contributed interactive applications and demonstrations to public-engagement events, supporting a reported 30% increase in participation.

Full Stack Developer, Part-Time

IMaR Technology Gateway | *2020 – 2021*

- Architected backend RESTful API services for a clinical trial Serious Adverse Event (SAE) application, building secure PHP Laravel 8.0 endpoints, structured MySQL schemas, and live SMS/alert broadcasting engines utilising Pusher.
- Collaborated directly with founder Wayne McMahon to engineer cross-platform patient-to-clinician alert interfaces using React, React Native, Leaflet.js, and Bootstrap 4, supporting startup validation within the Enterprise Ireland New Frontiers entrepreneurship programme.
- Optimized codebase maintainability by enforcing Git/GitHub version-control best practices, streamlining engineering collaboration workflow across shared software repositories.

Graduate R&D Software Intern

IMaR Technology Gateway | *June 2019 – October 2019*

- Automated hourly crane-metrics analysis for Liebherr Container Cranes Ltd by building a Python tool to process 383 MB of raw telemetry data (250k rows across 40 parameters), streamlining data pipelines to support preventative maintenance.
- Accelerated team onboarding and knowledge transfer by authoring detailed technical documentation of workflows and system architectures in an internal wiki.

Selected Projects

Volumetric Livestock Liveweight Estimation | Industry Research Project, IMaR Technology Gateway | *2025*
Python, 3D ToF Imaging, Computer Vision, OpenCV, Point Cloud

- Developed and evaluated a 3D computer-vision workflow for livestock liveweight estimation using adaptive background subtraction and point cloud reconstruction, while systematically documenting architectural constraints in occlusion, depth completeness, and segmentation.

Online Portfolio | Next.js, Tailwind CSS, EmailJS, CircleCI, Vercel | *2022*
Live site: dejongyeong.com | Repository**:** github.com/dejongyeong/v2

- Developed and deployed a responsive portfolio website to showcase software projects and professional experience, with automated CI/CD deployment workflows.

COVID-19 Data Tracker | React, Chart.js, React Simple Maps, Material UI, Heroku, CircleCI | *2021*
Live site: cov2-tracker.netlify.app | Repository: github.com/dejongyeong/covid-19-tracker

- Built an interactive data visualisation application to map global and country-specific pandemic metrics, integrating the Open Disease Data API with automated CI/CD deployment workflows.

Education

Master by Research in Engineering | Munster Technological University, Kerry | *2019 – 2022*
Development of Multi-Sensor Fusion Perception Systems for Obstacle Detection in Container Terminals

- Published multiple peer-reviewed papers, highlighting an autonomous sensor fusion review in the MDPI Sensors Journal with 1,600+ citations and 120k+ views utilising Python for analysis with academic exposure to ROS and C++.
- Collaborated across multidisciplinary teams to design physical components and delivered bi-weekly progress reports to industrial partners.

BSc (Hons) in Computing with Software Development | Institute of Technology Tralee | *2018 – 2019*
First Class Honours | Repository: github.com/dejongyeong/sentiment-analysis

- Engineered a hybrid sentiment analysis model using Python and Scikit-Learn to parse 400,000 Amazon reviews, achieving 81.2% accuracy to automate business brand reputation evaluation.

Achievements & Certifications

Academic Honors & Publications

- **Most Influential Paper Award (2021 – 2026)**: Selected by the Editor-in-Chief as one of only 25 review papers globally from a 25-year archive to celebrate the silver anniversary of the MDPI Sensors Journal, recognizing *"Sensor and Sensor Fusion Technology in Autonomous Vehicles: A Review"* for landmark research impact.
- **Best Paper Award (2023) & Editor's Choice Selection:** Awarded lead-author honours for the same publication (DOI: 10.3390/s21062140), which has driven outstanding global engagement, accumulating over 124,000 full-text views and 1,600+ citations to date.

Professional Certificates

- **Claude 101** | *Anthropic Education* | 2026
- **Research Integrity: Concise (Core Course)** | *Epigeum Academic Framework* | 2022
- **Microsoft Office Specialist 2010** | *Microsoft & Certiport* | 2013

Volunteering Experience

Workshop Instructor | Kerry Science Festival @ Munster Technological University | *2021*

- Delivered the "Let's code with Scratch" online workshop via Microsoft Teams to 34 primary school students in collaboration with internal research staff.

Program Coordinator | Institute of Technology Tralee / Trinity College Dublin Access21 Initiative | *2021*

- Guided secondary students through structured debates on climate change and emerging technologies, fostering collaborative critical thinking and active group engagement.

Selected Publications

- Yeong, D. J., Panduru, K., & Walsh, J. (2025). Exploring the Unseen: A Survey of Multi-Sensor Fusion and the Role of Explainable AI (XAI) in Autonomous Vehicles. *MDPI Sensors Journal*. DOI: 10.3390/s25030856
- Yeong, D. J., Panduru, K., & Walsh, J. (2024). Building Trustworthy Autonomous Vehicles: The Role of Multi-Sensor Fusion and Explainable AI (xAI) in On-road and Off-road Scenarios. *10th International Conference on Sensors and Electronics Instrumentation Advances (SEIA).* IFSA. DOI: 10.13140/RG.2.2.36670.50243.
- Yeong, D. J., Panduru, K., & Walsh, J. (2024). Smart Agriculture: Software Platform for Telematics Monitoring in Farm Machinery. *35th Irish Signals and System Conference (ISSC).* IEEE. DOI: 10.1109/ISSC61953.2024.10602840